Certified reliability for any routing solver: screening buffered plans against travel-time uncertainty, with evidence from real delivery routes

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Word count: about 9,400 words from the abstract to the end of the declarations, including tables and captions, excluding references.

Acknowledgements. The author thanks the researchers who answered methodological questions by e-mail during the project.
Disclosure of interest. The author reports there are no competing interests to declare.
Funding. This research received no external funding.
Data availability statement. All code, instances, records and analysis scripts are openly available in the archive cited in the manuscript (Zenodo, doi:10.5281/zenodo.23265196). The Amazon route sample is redistributed under its CC BY-NC 4.0 licence; the New York taxi records are public and are not copied.
Author contributions (CRediT). Suhan Ali: conceptualization, methodology, software, validation, formal analysis, investigation, data curation, visualization, writing – original draft, writing – review and editing.
Use of generative AI. Anthropic's Claude and OpenAI's ChatGPT were used to write and debug the simulation and analysis code, to review that code, to run and check the statistical analyses, to search the literature, and to draft and revise the text of the manuscript, including its wording. The research question, the study design and the confirmatory protocols were the author's. Every experiment was run from the archived code, every reported number is regenerated from the archived records, and the author takes full responsibility for the content. **Biographical note.** Suhan Ali is an independent researcher

based in Lahore, Pakistan, working on vehicle routing and delivery planning under travel-time uncertainty. Alongside this research, Suhan is developing RouteOps, a route-planning system built on Google OR-Tools.